

VIKASH KUMAR

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RESEARCH INTERESTS

Natural Language Processing, Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Information Retrieval, Machine Learning, AI for Knowledge Management.

EDUCATION

Balochistan University of Information Technology, Engineering and Management Sciences

Quetta, Pakistan

Bachelor of Science in Software Engineering, CGPA: **3.69/4.0**

Aug 2014 - Aug 2018

New Mexico Highlands University (NMHU)

New Mexico, USA

Semester Exchange Program, Computer Science, GPA: **3.75/4.0**

Aug 2017 - Dec 2017

AWARDS & HONORS

- **2nd Runner-Up** — Ignite AI Wrapper Competition 2025, GovTech Category — Awarded PKR 500,000 for AI-Powered Archival Management System (AI-AMS).
- **Global UGRAD Program Scholar** — Selected for a semester-long academic exchange at New Mexico Highlands University, USA, 2017.
- **2nd Prize** — **Balochistan Technology Award 2019**, University of Balochistan — Recognized for the IoT Water Quality Monitoring Kit.
- **PM National ICT Scholarship** — Selected among the top 500 recipients from approximately 50,000 applicants.

RESEARCH & TECHNICAL EXPERIENCE

Research Associate (ICT), Center of Excellence for Results Based Management, UoB

May 2020 – Dec 2021

- Designed and developed the Archival Management System (AMS) using Laravel, Vue.js, MySQL, and Elasticsearch to digitize, index, and retrieve thousands of PSDP reports spanning 1995–2020.
- Developed an HRMIS for the Health Department, supporting the digitization of personnel records for approximately 40,000 employees, including data validation, verification, and structured record management.
- Led a 40-member data-entry team and coordinated technical implementation, data quality, and stakeholder requirements; the archival platform later evolved into the AI-Powered Archival Management System (AI-AMS) with semantic retrieval and AI-assisted document intelligence.

Research Assistant, Electronic Lab, BUITEMS

Jan 2018 – Dec 2018

- Supported research fellows in multiple projects involving robotics equipment and experimental systems.
- Provided technical support for research activities and laboratory experimentation.
- Assisted with robotics workshops and student research activities.

RESEARCH & APPLIED AI SYSTEMS

AI-Powered Archival Management System (AI-AMS)

Research & AI Project — Evolved from AMS developed at CERBM

- Transformed an existing digital archival platform into an AI-assisted knowledge system for organizing, searching, and exploring government and institutional records.
- Developed NLP and information-retrieval workflows for document classification, metadata extraction, semantic search, and RAG-based access, enabling natural-language retrieval of archival information.
- Designed a human-in-the-loop verification workflow for AI-generated outputs, supporting traceability and reliable document processing; the project won 2nd Runner-Up in the GovTech category of the Ignite AI Wrapper Competition 2025, with a PKR 500,000 award.

Personalized Learning System for Exam Preparation . [GitHub](#)

[Llama 3 Hackathon](#) | *Python, Streamlit, RAG, FAISS, SentenceTransformers*

- Built a RAG-based learning assistant for SAT, ACT, IELTS, and TOEFL preparation using uploaded study materials as the knowledge source.
- Designed the retrieval pipeline using SentenceTransformer embeddings and FAISS semantic search, then grounded Llama-generated answers in retrieved study content.
- Implemented persistent user profiles, chat history, document ingestion, **retrieval, and response generation** in Streamlit to support personalized and continuous learning interactions.

AI Space Data Explorer · [GitHub](#)

[lablab.ai hackathon](#) | Python, LangGraph, Llama-3.3-70B, NASA APIs, Streamlit, Plotly

- Contributed to the development of an AI-powered application for conversational exploration of NASA space data.
- Contributed the Astronomy Picture of the Day (APOD) functionality, integrating NASA's APOD API into the application.
- Worked with a team to integrate LLM-based interaction, NASA data sources, and interactive data visualization into a Streamlit-based application.

DeepDiet — AI-Powered Nutrition Guide · [GitHub](#)

[lablab.ai hackathon](#) | Python, Streamlit, DeepSeek-R1, YOLOv10, USDA API

- Built an AI-powered nutrition assistant that generates personalized meal plans based on users' dietary goals and provides food-specific nutritional insights.
- Integrated 2 AI components, DeepSeek-R1 for natural-language meal planning and YOLOv10 for image-based food recognition, enabling users to interact through both text and uploaded food images.
- Connected detected food items to the USDA nutrition database and delivered the complete workflow in Streamlit, combining meal planning, food recognition, and nutritional analysis in a single application.

NLP-Based Author Identification Using Naive Bayes · [GitHub](#)

Python | Natural Language Processing | Machine Learning

- Built an authorship-classification system to identify unseen text using 10+ linguistic and Part-of-Speech features extracted from works by two authors.
- Trained and evaluated a Naive Bayes classifier using a 70/30 train-test split across three shuffled experimental runs to test robustness and consistency.
- Measured classification performance using individual and average accuracy across multiple runs, demonstrating the effectiveness of linguistic features for authorship prediction.

TEACHING EXPERIENCE

Lecturer (BPS-17) — Govt. Postgraduate Science College | Quetta, Pakistan

Dec 2021 – Present

- Teach undergraduate courses including Programming Fundamentals, Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, and Artificial Intelligence.
- Design and deliver lectures, laboratory exercises, programming assignments, and assessments for undergraduate Computer Science students.
- Supervised and mentored 8 final-year capstone projects, supporting students throughout project development and evaluation.

PROFESSIONAL EXPERIENCE

IT Consultant — Provincial Assembly of Balochistan (Project-Based) | Quetta, Pakistan

Dec 2022 – Feb 2025

- Developed the Public Accounts Committee Information System (PACIS) to digitize and manage audit and financial records used in legislative oversight.
- Contributed to the development of institutional digital information systems for the Provincial Assembly.

Software Engineer — Government Innovation Lab, UoB | Quetta, Pakistan

July 2019 – April 2020

- Developed an IoT-based Water Quality Monitoring System integrating pH, turbidity, TDS, and temperature sensors with cloud-based data collection for real-time monitoring.
- Built web and mobile dashboards for real-time visualization and analysis of sensor data, while improving measurement reliability; the project won 2nd Prize at the Balochistan Technology Award 2019.

Junior Software Engineer — AUM Technologies (PVT) LTD | Quetta, Pakistan

Jan 2019 – June 2019

- Developed and tested web applications and contributed across the software development lifecycle.

TECHNICAL SKILLS

- **AI & NLP:** Natural Language Processing, Machine Learning, LLMs, RAG, Information Retrieval, Semantic Search, Text Classification
- **AI/ML Tools:** SentenceTransformers, FAISS, LangGraph, OpenAI API, Pandas
- **Programming & Data:** Python, SQL, PHP, JavaScript, C# | MySQL, SQL Server, MongoDB, Elasticsearch
- **Development & Visualization:** Git, GitHub, Docker, Streamlit, Plotly

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Generative AI with Large Language Models** — [DeepLearning.AI, Coursera, 2024](#)
- **Google Cybersecurity Professional Certificate** — [Google](#)